higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION: BOILERMAKING
NQF LEVEL 4

(6031014)

22 November 2019 (X-Paper)
09:00–12:00

Calculators may be used.

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Use only blue or black ink.
6. Use 'g' as 9,8 ms^{-2}.
7. Write neatly and legibly.

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SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1–1.7) in the ANSWER BOOK.

1.1 Which description describes a double-bevel weld symbol?

A Symbols have their legs at 60° to each other.
B Symbol is a 45° V with its apex to the reference line.
C Small isosceles triangle drawn on the reference line.
D Symbol is a 60° V with its apex to the reference line.

1.2 Which description explains the meaning of the weld throat thickness?

A Distance measured across the face of the weld.
B The thickness of the welded plate.
C The total length of the weld.
D Thickness of weld metal measured along a line passing through the root.

1.3 What is the function of a saddle on a pressure vessel?

A Used to climb onto vessel.
B Steel or concrete supports for horizontal vessels.
C Used as a platform during inspection.
D To monitor pressure of the vessel.

1.4 Calculate the mass of an I-section with the following size:

178 mm × 102 mm × 22 kg/m with a length of 4 m.

A 102 kg
B 178 kg
C 88 kg
D 108 kg

1.5 Which tool will be used when marking off extra-large arcs on a metal sheet?

A Scriber
B Trammels
C Divider
D Chalk line

1.6 Which formula will be used to calculate the area of a circle?

A πr^2
B πD
C $2\pi r$
D $2\pi d$

1.7 Which tool will be used to secure two plates together temporarily for marking off or drilling?

A V-block with stirrup clamp
B G-clamp
C Pliers
D Weights

(7 × 1) **[7]**

QUESTION 2

Choose the CNC 'G' code in COLUMN B that matches a description from COLUMN A. Write only the letter (A–L) next to the question number (2.1–2.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1	Workpiece co-ordinate system	A	G0
2.2	X-Y plane for circular interpolation	B	G1
2.3	Programming in inches	C	G2
2.4	Moves the axis at rapid speed	D	G3
2.5	Exact stop check	E	G4
2.6	Programming in mm	F	G9
2.7	Z-X plane for circular interpolation	G	G17
2.8	Moves the axis at the specified feed rate	H	G18
2.9	The dwell or pause for a certain amount of time	I	G19
		J	G20
2.10	Y-Z plane for circular interpolation	K	G21
		L	G54

(10 × 1) **[10]**

QUESTION 3

The following abbreviations are related to a term regards fabrication.

State the meaning of each abbreviation.

3.1 Pressure vessels – 'G'

3.2 Structural steel – 'bev'

3.3 Pipe work – 'ELL'

3.4 CNC – 'R'

(4 × 1) **[4]**

TOTAL SECTION A: 21

SECTION B

QUESTION 4

4.1 Draw a neat free hand drawing of a V-groove weld and indicate the following: (1)

4.1.1 V-groove weld width (1)

4.1.2 Depth of weld (1)

4.1.3 Write down the formula for the *bevelled area* of the V-groove (2)

4.2 A combination set has three heads.

What can you do with of each of the THREE heads in conjunction with the ruler? (3 × 2) (6)

4.3 Describe the THREE steps of using a back-mark gauge. (3)

4.4 Calculate the length of plate required to fabricate a cylinder with an outside diameter of 1 450 mm and plate thickness of 12 mm. (4)

4.5 There are situations where liquids cannot be used for pressure testing.

Name TWO gasses that can be used for the pressure testing of vessels. (2)

[20]

QUESTION 5

5.1 List THREE types of structural drawings that you need to know before you can learn to examine drawings. (3)

5.2 Explain what is meant by the following drawing terms:

5.2.1 Isometric drawing

5.2.2 Orthographic projection

(2 × 2) (4)

5.3 When you do periodic inspection of hoists, what must you check? (6 × 2) (12)

[19]

QUESTION 6

6.1 List THREE points of information that could be found on a spool drawing for pipework. (3)

6.2 The pipefitter's square is based on the same principle as the framing square.

Describe the main difference between the *pipefitter's square* and the *framing square.* (2)

6.3 Calculate the pressure exerted at the base of a vertical column with a length of 60 m when it is filled with sea water with a density of 1 100 kg/m^3. (5)

HINT: Pressure = ρgh.

6.4 Calculate the area of a circle, if the diameter is 120 mm. (5)

6.5 State THREE points to be considered when selecting the correct measuring and checking tools for a pipework assembly job. (3 × 2) (6)

[21]

QUESTION 7

7.1 Define the term *toolpath*, regards to CNC work. (2)

7.2 You need to know all machine positions because they are important for programming a CNC machine.

Of the four machine positions, which THREE are set by the manufacturer? (3)

7.3 List SEVEN features that must be identified on an engineering drawing. (7)

7.4 Explain the difference between *absolute* and *incremental* movement. (2 × 2) (4)

7.5 List THREE factors that determine the amount of material that can be removed during machining. (3)

[19]

TOTAL SECTION B: 79
GRAND TOTAL: 100

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